Design and Analysis of a Split-Plot Experiment to Assess the Effects of Educational Intervention and Support Strategies on Adolescent Smoking

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Background

Adolescent smoking is a major public health concern because smoking during this developmental period is associated with impaired lung development, increased risk of respiratory infection, and elevated risk of lung cancer (Park, 2011). Approximately 90% of smokers begin before the age of 18, and earlier initiation substantially increases the difficulty of smoking cessation later in life (Park, 2011). School-based interventions are therefore important targets for prevention efforts, as schools represent one of the primary environments influencing adolescent smoking behavior.

This study evaluated the effects of school-based educational and support interventions on cigarette smoking among eighth-grade students. The educational intervention consisted of either no presentation or a two-hour presentation featuring testimonials from former smokers. The support strategy consisted of one of four approaches: no support, a follow-up video, a question-and-answer session with a former smoker, or both the video and question-and-answer session combined.

The primary objective of the study was to determine whether the educational intervention and support strategy affected average cigarette consumption among students.

Methods

Study Design

A split-plot experimental design was used to evaluate the effects of an educational intervention and support strategies on cigarette smoking among eighth-grade students, with districts serving as blocks (Barkauskas, 2026, Chs. 11, 14). Based on a priori power calculations (Barkauskas, 2026, Chs. 4, 8, 10, 14), a minimum of 7 districts were required to achieve at least 90% power for both hypothesis tests at $\alpha = 0.05$. Eleven districts were recruited for the definitive trial. Within each district, two schools were randomly assigned to one of two educational intervention levels: no presentation or a two-hour presentation featuring testimonials from former smokers. Within each school, four homerooms containing 8–12 students were randomly assigned to one of four support strategies: no support, a follow-up video, a one-hour question-and-answer session with a former smoker, or both the video and question-and-answer session combined.

The educational intervention constituted the whole-plot factor, while support strategy constituted the subplot factor (Barkauskas, 2026, Ch. 14). The response variable was the average number of cigarettes smoked per student during the week preceding a survey administered one month after treatment assignment. Because of privacy concerns related to collecting individual smoking behavior from minors, only the total number of cigarettes smoked within each homeroom and the number of students in each homeroom were recorded. The outcome analyzed for each homeroom was therefore calculated as the total number of cigarettes smoked divided by the number of students in the homeroom.

Hypotheses

The first null hypothesis tested whether the educational intervention affected cigarette smoking behavior among students, where α denotes the educational intervention:

$$H_0^{(1)} : \alpha_1 = \alpha_2 = 0$$

The second null hypothesis tested whether support strategy affected cigarette smoking behavior, where γ denotes the support strategy:

$$H_0^{(2)} : \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0$$

Statistical Analysis

Statistical analyses were conducted using split-plot analysis of variance (ANOVA).

$$Y_{ijk} = \mu + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + b_k + \varepsilon_{ijk}$$

The fitted model included fixed effects for educational intervention, support strategy, and their interaction, while district was treated as a blocking factor. In the model, Yijk denotes the average number of cigarettes smoked per student for the jth support strategy within the ith educational intervention in the kth district, μ represents the overall mean smoking rate, αi represents the fixed effect of educational intervention, γj represents the fixed effect of support strategy, (αγ)ij represents the interaction effect between educational intervention and support strategy, bk represents the random effect of district, and εijk represents the subplot residual error term. Educational intervention effects were tested against the district-by-educational-intervention mean square, whereas support strategy and interaction effects were tested against the subplot residual mean square. The response variable analyzed was the average number of cigarettes smoked per student within each homeroom. Treatment variables were analyzed as categorical factors, and statistical significance was assessed at the 0.05 level.

Ninety-five percent confidence intervals were constructed for the educational intervention effects, support strategy effects, and pairwise differences between support strategies using the appropriate expected mean square for each comparison. Pairwise comparisons among support strategies were used to identify which intervention combinations differed significantly in average cigarette consumption.

Software

All analyses were performed in R using the packages readxl, dplyr, ggplot2, and tidyverse. R version 4.4.0 was used.

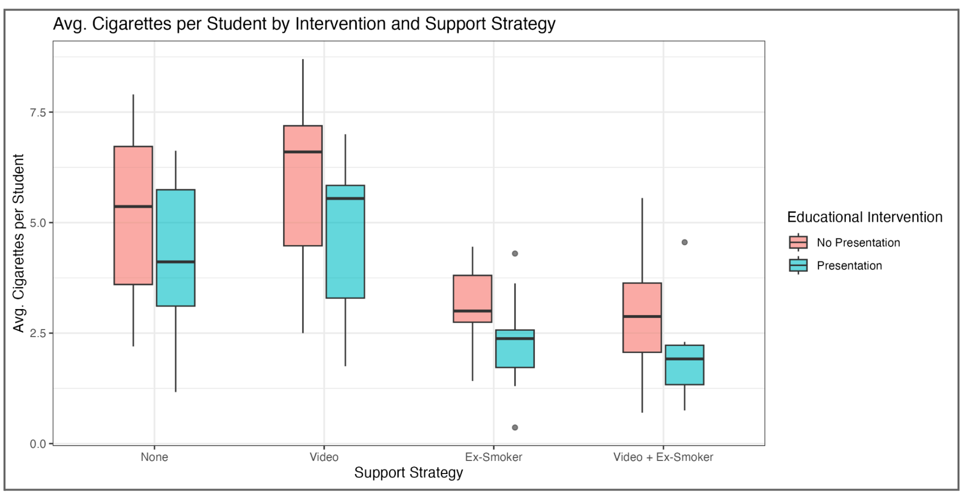
Results

Table 1. Average Cigarettes Smoked per Student by Treatment Group; Mean (SD)

Support Strategy	No Presentation	Presentation	Marginal Mean
None	5.24 (1.88)	4.29 (1.75)	**4.77**
Video	5.94 (1.94)	4.75 (1.77)	**5.34**
Ex-Smoker	3.13 (0.93)	2.31 (1.07)	**2.72**
Video + Ex-Smoker	3.00 (1.41)	1.97 (0.99)	**2.49**
Marginal Mean	**4.33**	**3.33**	

Prior to formal hypothesis testing, descriptive statistics were examined for the average number of cigarettes smoked per student across treatment groups (Table 1). Across both intervention groups, students in the Ex-Smoker and Video + Ex-Smoker support strategy conditions had lower average smoking rates compared to those in the None and Video conditions. Students who received the presentation had lower average smoking rates than those who did not, across all four support strategies.

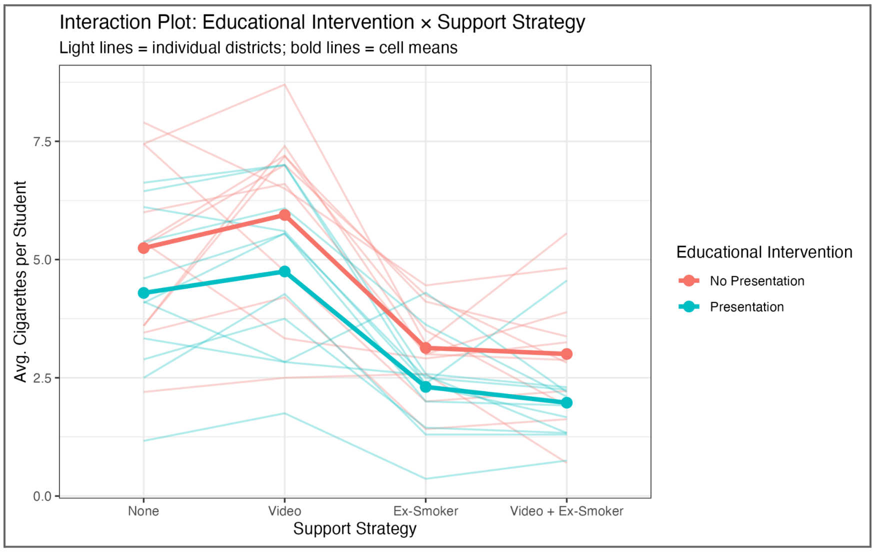
Figure 1



Homerooms assigned to the No Presentation condition showed consistently higher median smoking rates than those assigned to the Presentation condition across all four support strategies (Figure 1).

Within both intervention groups, the Ex-Smoker and Video + Ex-Smoker strategies were associated with notably lower median smoking rates and reduced spread compared to the None and Video strategies, with the Video strategy showing little reduction relative to no support.

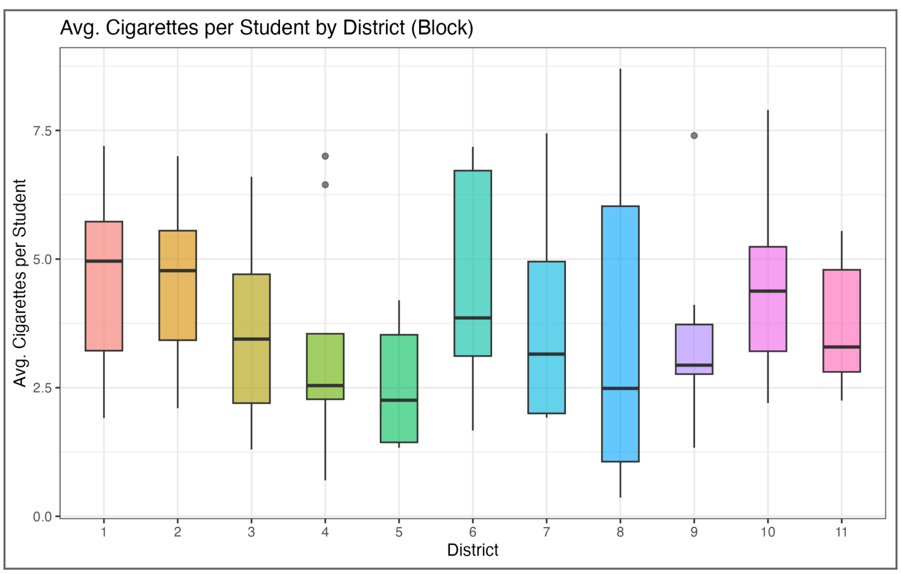
Figure 2



The interaction plot (Figure 2) revealed that the two mean lines were nearly parallel across all four support strategy levels, suggesting little evidence of an interaction between educational intervention and support strategy.

Both groups show a similar pattern: average smoking rates were highest under the None and Video conditions and decreased substantially under the Ex-Smoker and Video + Ex-Smoker conditions. The Presentation group consistently showed lower average smoking rates than the No Presentation group across all support strategies, providing preliminary evidence of a main effect of educational intervention.

Figure 3

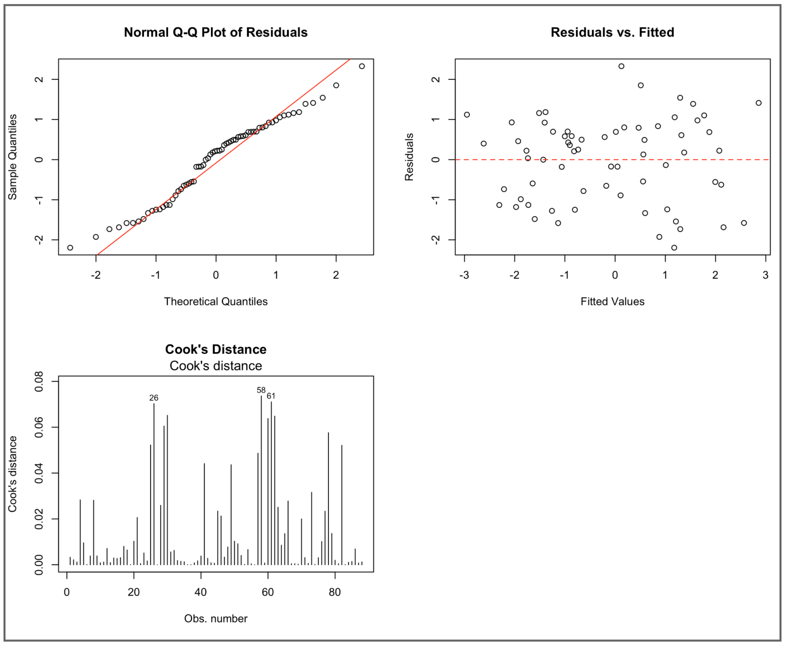


There is considerable variability in average smoking rates across the 11 districts, with median rates ranging from approximately 2.3 in District 9 to nearly 4.0 in Districts 1 and 2 (Figure 3).

This between-district variability confirms that blocking on district was an appropriate design choice, as districts differ meaningfully in their baseline smoking rates. A few mild outliers are visible in Districts 4, 9, and 8, though none appear extreme enough to raise serious concern.

Prior to hypothesis testing, the split-plot ANOVA model was fitted and model assumptions were assessed visually. The normal Q-Q plot, residuals-vs.-fitted plot, and Cook's distance are displayed in Figure 4.

Figure 4



The normal Q-Q plot showed that the residuals followed the reference line closely in the central region, with mild departures in the tails, suggesting the normality assumption was reasonably satisfied. The residuals-vs.-fitted plot showed no obvious pattern or systematic trend, with residuals scattered approximately symmetrically around zero across the range of fitted values, supporting the homoscedasticity assumption. Cook's distance values were well below the conventional threshold of 1 for all observations. However, observations 26, 58, and 61 exceeded the $4/N = 4/88 \approx 0.045$ threshold, with Cook's distances of approximately 0.07. These observations were examined but not removed, as their influence was not considered practically meaningful and all values remained far below the stricter threshold of 1. Overall, the diagnostic plots did not reveal any serious violations of model assumptions, and no transformations or corrective measures were deemed necessary.

Table 2. Split-Plot ANOVA for Average Cigarettes Smoked per Student

Stratum	Source	df	SS	MS	F	p-value
Block	District	10	34.51	3.451	—	—
Whole Plot	Educational Intervention	1	21.97	21.971	2.733	0.129
	Residual	10	80.39	8.039		
Subplot	Support Strategy	3	136.61	45.54	38.988	<0.001
	EDUC × SUPP	3	0.40	0.13	0.114	0.952
	Residual	60	70.08	1.17		

The interaction between educational intervention and support strategy was not statistically significant ($F(3,60) = 0.114$, $p = 0.952$), indicating that the effect of support strategy on average cigarette consumption did not differ by educational intervention level. The educational intervention was not statistically significant ($F(1,10) = 2.733$, $p = 0.129$), providing insufficient evidence that the presentation reduced average smoking rates. Support strategy was highly statistically significant ($F(3,60) = 38.988$, $p < 0.001$), indicating strong evidence that at least one support strategy differed from the others in its effect on average cigarette consumption (Table 2).

Table 3. Point Estimates and 95% Confidence Intervals for Treatment Effects

Educational Intervention		
Effect	**Estimate**	**95% CI**
No Presentation (α_1)	0.499	(−0.174, 1.173)
Presentation (α_2)	−0.499	(−1.173, 0.174)
Support Strategy		
Effect	**Estimate**	**95% CI**
None (γ_1)	0.938	(0.539, 1.337)
Video (γ_2)	1.515	(1.116, 1.914)
Ex-Smoker (γ_3)	−1.111	(−1.510, −0.712)
Video + Ex-Smoker (γ_4)	−1.342	(−1.741, −0.943)

Estimates are in the sum-to-zero parametrization. CIs for educational intervention use the whole-plot error (df = 10); CIs for support strategy use the subplot error (df = 60).

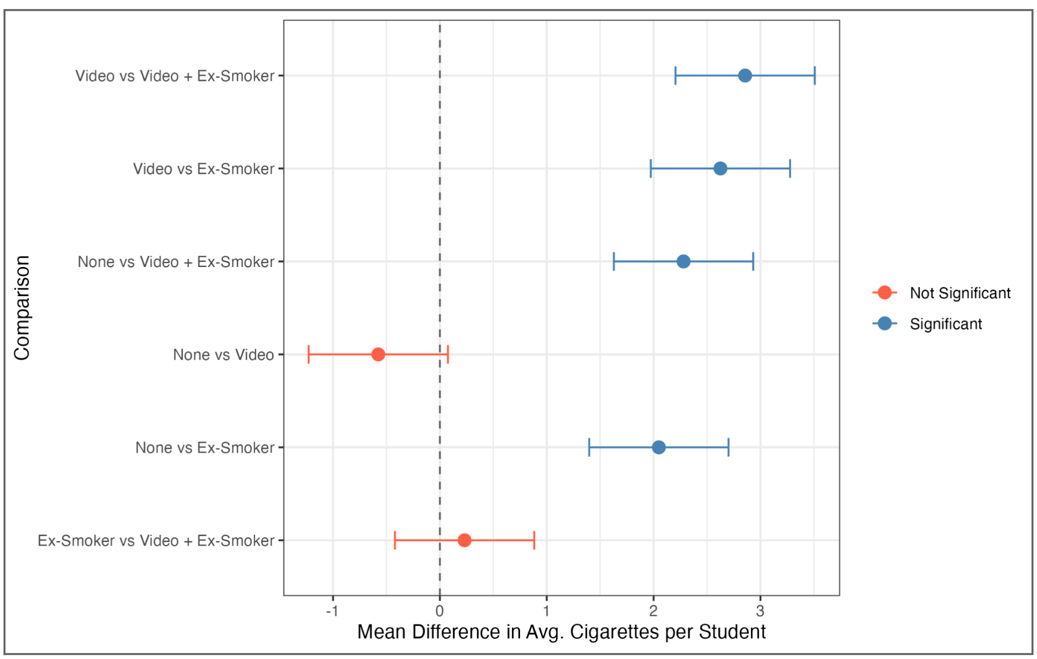
For the educational intervention, the No Presentation group had an estimated effect of 0.499 cigarettes above the grand mean, while the Presentation group had an estimated effect of −0.499 cigarettes below the grand mean (Table 3). However, both confidence intervals contained zero, consistent with the non-significant F-test ($p = 0.129$), indicating insufficient evidence of a meaningful educational intervention effect. For support strategy, the confidence intervals were notably narrower than those for educational intervention, reflecting the greater precision afforded by the subplot error term (df = 60) compared to the whole-plot error (df = 10). The None and Video conditions had confidence intervals entirely above zero (0.539, 1.337) and (1.116, 1.914), respectively (Table 3), while the Ex-Smoker and Video + Ex-Smoker conditions had confidence intervals entirely below zero (−1.510, −0.712) and (−1.741, −0.943), respectively (Table 3). These results are consistent with the highly significant support strategy F-test ($F(3,60) = 38.988$, $p < 0.001$), confirming that both Ex-Smoker and Video + Ex-Smoker strategies were associated with meaningfully lower average smoking rates relative to the grand mean, while the Video strategy alone was associated with higher rates. Given the statistically significant support strategy effect, pairwise comparisons were conducted to identify which strategies differed significantly from one another (Table 4).

Table 4. Pairwise Differences in Average Cigarettes Smoked per Student Between Support Strategies

Comparison	**Difference**	**95% CI**
None vs. Video	−0.576	(−1.228, 0.076)
None vs. Ex-Smoker	2.049	(1.397, 2.701)
None vs. Video + Ex-Smoker	2.280	(1.629, 2.932)
Video vs. Ex-Smoker	2.626	(1.974, 3.277)
Video vs. Video + Ex-Smoker	2.857	(2.205, 3.508)
Ex-Smoker vs. Video + Ex-Smoker	0.231	(−0.421, 0.883)

Differences are expressed as the first group minus the second group. CIs use the subplot error (df = 60). Confidence intervals not containing zero indicate a statistically significant difference at the 0.05 level.

Figure 5. Pairwise Differences in Average Cigarettes Smoked per Student Between Support Strategies



Pairwise comparisons among support strategies are presented in Table 4. The two within-group comparisons, None vs. Video (difference = −0.576, 95% CI: −1.228, 0.076) and Ex-Smoker vs. Video + Ex-Smoker (difference = 0.231, 95% CI: −0.421, 0.883), were not statistically significant, as both confidence intervals contained zero. In contrast, all comparisons between a strategy with no ex-smoker contact and a strategy with ex-smoker contact were statistically significant, with differences ranging from 2.049 to 2.857 cigarettes per student. This pattern suggests that direct contact with a former smoker was the active ingredient driving reductions in average cigarette consumption, regardless of whether it was combined with a video. Notably, the addition of a video to the ex-smoker session did not produce a statistically significant further reduction, suggesting that the video alone contributes little beyond what the ex-smoker contact already achieves. These findings are further illustrated in Figure 5.

Discussion

This study aimed to evaluate the effects of educational intervention and support strategy on average cigarette consumption among eighth-grade students using a split-plot experimental design. The results of this study indicate that support strategy is the primary driver of reduced cigarette consumption among eighth-grade students, with direct contact with a former smoker, either alone or combined with a video, associated with substantially lower average smoking rates compared to no support or video alone. Based on these findings, the recommended combination for future implementation is the two-hour educational presentation paired with the Video + Ex-Smoker support strategy. Although the educational intervention did not reach statistical significance, the Presentation group consistently showed lower average smoking rates across all support strategy conditions, and the non-significant result may partly reflect limited statistical power at the whole-plot level, where only 11 districts contributed to the test. The Video + Ex-Smoker strategy had the largest estimated reduction in smoking relative to the grand mean ($\gamma_4 = -1.342$), and while it did not differ significantly from the Ex-Smoker strategy alone, the combination trended lower and adds minimal additional burden relative to its potential benefit.

It is also worth noting that the video alone, without ex-smoker contact, showed no meaningful benefit over no support at all, suggesting that passive media-based interventions are insufficient on their own and that the interpersonal element of the ex-smoker session is likely the mechanism through which behavioral change occurs.

A key limitation of this study is that individual-level smoking data were unavailable due to privacy concerns, and the outcome analyzed was the homeroom-level average rather than individual student responses. This aggregation may have reduced sensitivity to detect true treatment effects, particularly for the educational intervention, as homeroom-level averaging obscures individual variation in smoking behavior. Additionally, the self-reported nature of cigarette consumption, even when aggregated, may be subject to social desirability bias, as students may underreport smoking behavior even in an anonymous group setting. The generalizability of these findings also depends on whether the 11 recruited districts are representative of the broader population of school districts in terms of demographic composition, baseline smoking rates, and school culture. Future studies with a larger number of districts would improve power for detecting whole-plot effects and provide more precise estimates of the educational intervention effect, and individual-level data collection, where ethically feasible, would strengthen the sensitivity of the analysis.

References

Barkauskas, D. (2026). *PM 513: Experimental Designs Course Notes*, Chapters 3, 4, 5, 6, 9, 14. Department of Population and Public Health Sciences, Keck School of Medicine, University of Southern California, Los Angeles, CA.

Park, S. H. (2011). *Smoking and adolescent health.* Korean Journal of Pediatrics, 54(10), 401–404. https://doi.org/10.3345/kjp.2011.54.10.401